fewer peaks and troughs—a string can embody less and less energy. But as we found in Chapter 4 in a different context, quantum mechanics tells us that this reasoning is not right. Like all vibrations or wavelike disturbances, quantum mechanics implies that they can exist only in discrete units. Roughly speaking, just as the money carried by a comrade in the warehouse is a whole number multiple of the monetary denomination with which he or she is entrusted, the energy embodied in a string vibrational pattern is a whole number multiple of a minimal energy denomination. In particular, this minimal energy denomination is proportional to the tension of the string (and it is also proportional to the number of peaks and troughs in the particular vibrational pattern), while the whole number multiple is determined by the amplitude of the vibrational pattern.

The key point for the present discussion is this: Since the minimal energy denominations are proportional to the string's tension, and since this tension is enormous, the fundamental minimal energies are, on the usual scales of elementary particle physics, similarly huge. They are multiples of what is known as the Planck energy. To get a sense of scale, if we translate the Planck energy into a mass using Einstein's famous conversion formula E = mc

2, they correspond to masses that are on the order of ten billion billion (1019) times that of a proton. This gargantuan mass—by elementary particle standards—is known as the Planck mass; it's about equal to the mass of a grain of dust or a collection of a million average bacteria. And so, the typical mass-equivalent of a vibrating loop in string theory is generally some whole number (1, 2, 3, …) times the Planck mass. Physicists often express this by saying that the "natural" or "typical" energy scale (and hence mass scale) of string theory is the Planck scale.

This raises a crucial question directly related to the goal of reproducing the particle properties in Tables 1.1 and 1.2: If the "natural" energy scale of string theory is some ten billion billion times that of a proton, how can it possibly account for the far-lighter particles—electrons, quarks, photons, and so on—making up the world around us?

The answer, once again, comes from quantum mechanics. The uncertainty principle ensures that nothing is ever perfectly at rest. All